DOMAINS: AI • WEB • CLOUD • OPEN INNOVATION

BUILD WITH भारत 5.0

NATIONAL LEVEL HACKATHON

TEAM NAME: Empty (Project: DrishtiFlow / दृष्टि-फ़्लो)

PROBLEM STATEMENT: DrishtiFlow — Vernacular No-Code AI Automation for Existing CCTV Cameras

TEAM MEMBERS: Farhan Mallik (Lead), Kartikey Bisht, Ankit Maurya, Piyush Kumar Gautam, Kaushendra Singh

COLLEGE: KCC Institute of Technology and Management, Greater Noida

PROBLEM STATEMENT

THE PASSIVE CCTV PROBLEM | Millions of Indian CCTV Cameras Only Record Footage for Post-Incident Review

Passive 'Dumb' CCTV: 99% Unmonitored
Cameras in mandis, MSMEs, hospitals & shops only store DVR video for review AFTER a loss.

Upgrade Cost: ₹25,000–₹1L+ CapEx
Smart AI NVRs, dedicated GPU servers, or 24/7 guards are unaffordable for small businesses.

Complex Dashboards: Zero Hindi UX
Legacy VMS tools require technical thresholds, coding, or English-only enterprise menus.

1. Mandi, Godown & Shop Blindspots
- **Late Discovery of Theft & Fire:** Godown and kirana owners discover night break-ins, smoke, or stray cattle damage only the next morning.
- **Alert Fatigue from Motion DVRs:** Basic motion sensors trigger constant false alarms from shadows, ceiling fans, or passing vehicles.

2. Unmonitored MSME & Hospital Safety
- **MSME Factory Floor Hazards:** Small factories lack automated real-time alerts when workers operate machinery without helmets or PPE.
- **Hospital OPD Triage Delays:** Crowded hospital waiting areas miss sudden patient falls or fainting spells when staff are occupied.

3. Hardware & Bandwidth Lock-In
- **Expensive Hardware Replacement:** Legacy AI camera vendors force merchants to replace working webcams and IP cameras.
- **Heavy Cloud Streaming Bills:** Uploading 24/7 video streams to cloud servers chokes Tier-2/3 internet and inflates monthly costs.

SOLUTION

DRISHTIFLOW (दृष्टि-फ़्लो) | Browser-Based 'Zapier for CCTV' Turning Any Camera into an Active AI Agent

PILLAR 1: VOICE-TO-WORKFLOW BUILDER

- **Speak Rules in Hindi / Hinglish:** Users say: 'Agar godown mein raat 10 baje ke baad koi dikhe ya dhuwan uthe, toh mujhe call karo'.
- **Auto-Generated Node Graph:** Gemini 2.5 compiles spoken vernacular commands into a visual Trigger → Reason → Act JSON graph.
- **Zero Extra Hardware Needed:** Runs in any modern web browser on top of existing USB webcams or IP CCTV cameras.

PILLAR 2: HYBRID EDGE + VLM ENGINE

- **Zero-Server Edge Filter:** In-browser vision (YOLOv8 + ONNX Runtime Web) detects motion, people, or objects locally with zero lag.
- **Smart Context Verification:** When triggered, Gemini Video & Groq verify real context (e.g., bending down vs. fainting in an OPD).
- **Near-Zero False Alarms:** Two-stage filtering keeps cloud bandwidth minimal while guaranteeing high-precision alerts.

PILLAR 3: REAL-WORLD ACTIONS & PRESETS

- **Automated Hindi AI Phone Call:** Places an instant outbound phone call in Hindi via VAPI to wake or alert the business owner.
- **WhatsApp Snapshot & Siren:** Sends an annotated WhatsApp photo proof and plays a loud regional voice warning on the local speaker.
- **Built-In Bharat Presets:** 1-click templates for Mandi/Godown Security, MSME Factory Safety, Hospital OPD & Kirana Stores.

FLOW OF SOLUTION

4-STEP AUTOMATION LOOP | Voice-to-Workflow → Zero-Server Edge Filter → Smart Context Check → Action

STEP 1: VOICE-TO-WORKFLOW

Gemini 2.5 Compiler

User speaks a command in Hindi/Hinglish/English; Gemini 2.5 converts it into a drag-and-drop Trigger → Reason → Act JSON node graph.

STEP 2: EDGE VISION FILTER

YOLOv8 + ONNX Web

Lightweight in-browser vision (YOLOv8 + ONNX Runtime Web) detects motion, people, or objects locally with zero cloud lag.

STEP 3: SMART CONTEXT CHECK

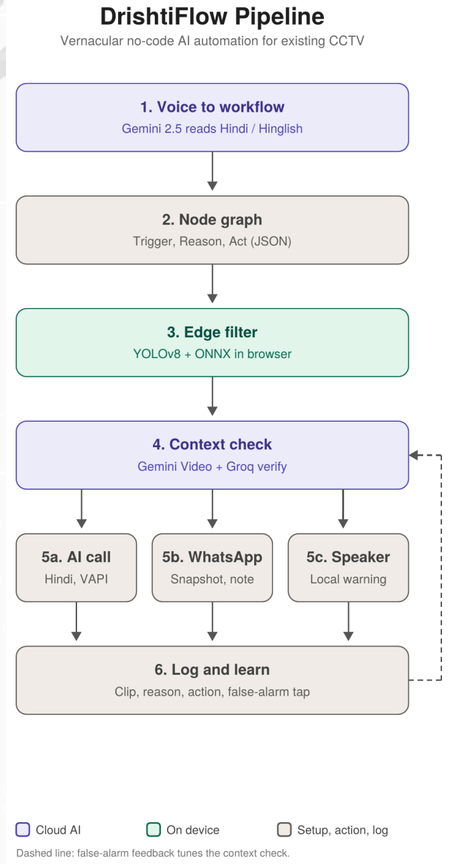
Gemini Video + Groq

When triggered, Gemini Video and Groq verify actual context (e.g., distinguishing bending down vs. fainting in a hospital OPD).

STEP 4: INSTANT ACTION

VAPI + WhatsApp + TTS

Automatically places an AI phone call in Hindi (VAPI), sends a WhatsApp snapshot, or plays a loud regional voice warning on-site.



TECH STACK

FULL-STACK ARCHITECTURE | Gemini 2.5 & Video + YOLOv8 ONNX Web + Groq LPU + VAPI Voice + Next.js Canvas

1. In-Browser Edge Vision (Zero Lag)

- **YOLOv8 + ONNX Runtime Web:** Client-side WebGPU/WASM object & pose detection at 30 FPS.
- **Zero Extra Hardware:** Works with any USB webcam, laptop cam, or RTSP IP CCTV feed.
- **Local Frame Gating:** Filters 99% of idle frames locally before calling cloud VLMs.

2. Multimodal AI & Voice Engine

- **Gemini 2.5 (Voice-to-Flow):** Compiles Hindi/Hinglish speech into Trigger-Reason-Act JSON.
- **Gemini Video (Context Check):** Verifies keyframe bursts (e.g., bending down vs. fainting).
- **Groq LPU Inference:** Sub-200ms rule reasoning & JSON payload orchestration.

3. No-Code Workflow UI & Backend

- **Next.js 15 + React Flow Canvas:** Visual Trigger → Reason → Act node editor with live bounding-box camera preview.
- **TypeScript + Tailwind CSS:** Bilingual Hindi/English dashboard with 1-click Bharat Preset selector.
- **FastAPI + Supabase / Webhooks:** Manages preset templates, incident snapshot logs, and webhook dispatch.

4. Action Integrations & 5-Member Split

- **VAPI Conversational Voice AI:** Places immediate outbound Hindi phone calls explaining the alert to the owner.
- **WhatsApp Cloud API + Local TTS:** Sends annotated snapshot alerts on WhatsApp & blasts regional audio warnings.
- **5-Member Role Split:** Farhan (Gemini/Lead), Kartikey (ONNX/YOLO), Ankit (Groq/Video), Piyush (UI), Kaushendra (VAPI).

UNIQUE SELLING PROPOSITION (USP)

CORE INNOVATION & USP | Zero Extra Hardware, Vernacular Voice Programming & Two-Stage Verification

USP 1: Vernacular Voice-to-Workflow ('Zapier for CCTV')

No coding or complex menus: users speak a rule in Hindi or Hinglish ('Agar godown mein raat 10 baje ke baad koi dikhe...'), and Gemini 2.5 builds the visual Trigger → Reason → Act workflow.

USP 2: 100% Browser-Native Edge Vision (₹0 Hardware)

Runs YOLOv8 via ONNX Runtime Web directly in the browser on existing PCs and CCTV feeds—eliminating ₹40,000+ AI NVR boxes and cutting 95%+ of continuous cloud streaming bandwidth.

USP 3: Two-Stage Edge + Gemini Video Context Check

Solves the false-alarm crisis of traditional motion/YOLO systems: YOLOv8 catches fast motion locally, while Gemini Video & Groq verify true semantics (e.g., bending down vs. fainting in an OPD).

USP 4: Active Real-World Intervention & Bharat Presets

Instead of passive dashboard logs, DrishtiFlow calls the owner's phone in Hindi via VAPI, sends a WhatsApp snapshot, and plays a loud warning over the local speaker—with 1-click Bharat templates.

FEASIBILITY & COMPETITORS

COMPETITOR BENCHMARK & FEASIBILITY | Enterprise AI Vision Power on Zero-Cost Existing Hardware for Bharat

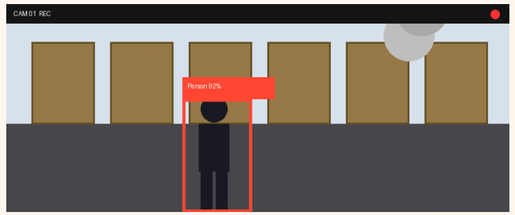
Capability / Feature	Legacy DVR / CCTV	Enterprise AI NVR	Cloud VMS (Cameron)	DrishtiFlow
Extra Hardware Cost	₹0 (Passive)	₹40k–₹1L+ Box	Cloud GPU Fees	**₹0 (In-Browser)**
Vernacular Voice Setup	No	No (Tech UI)	English-only	**Hindi / Hinglish**
Edge + VLM Context	Motion only	Fixed rules	Cloud-heavy	**YOLOv8 + Gemini**
Hindi AI Phone Call	No	App push only	Webhook only	**Yes (VAPI + WA)**
1-Click Bharat Presets	No	No	Generic	**Mandi/OPD/MSME**

FEASIBILITY & RISK MITIGATION

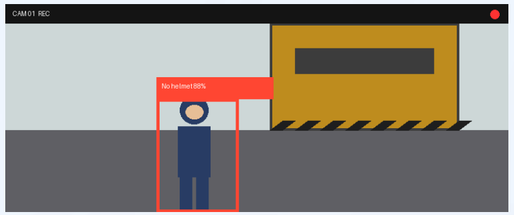
- **1. Zero Bandwidth Bottleneck:** YOLOv8 + ONNX Runtime Web filters 99% of frames locally; Gemini Video is called only when an edge trigger fires.
- **2. Universal Camera Compatibility:** Connects to any laptop webcam, USB camera, or existing RTSP/ONVIF IP camera via lightweight WebRTC bridge.
- **3. Finale Stage Demo Ready:** Speak a Hindi rule live at Microsoft Gurugram, simulate a fall/intrusion on webcam, and receive a live VAPI Hindi call in <5s.

EXPECTED IMPACT

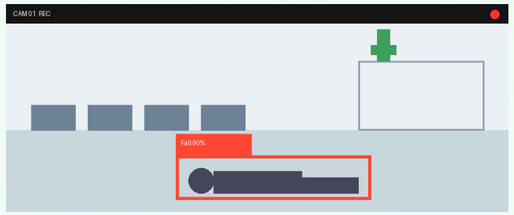
EXPECTED IMPACT & BHARAT PRESETS | 1-Click Protection Across Mandis, MSMEs, Hospital OPDs & Kirana Stores



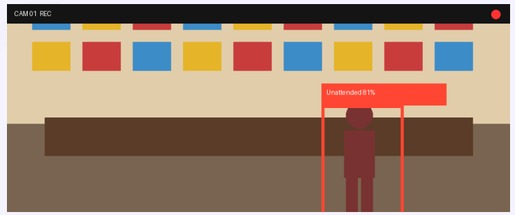
Mandi & Godown Security
Detects night intrusion, smoke/fire, or stray cattle in grain storage.



MSME Factory Safety
Flags missing helmets or PPE near active hazardous machinery.



Hospital OPD Triage
Distinguishes patient falls or fainting from bending down.



Kirana Counter Alerts
Spots unattended customers or counter theft in real time.

1. ₹0 Hardware CapEx for MSMEs
- **63M+ Indian MSMEs & Shops:** Upgrades existing 'dumb' CCTV cameras into active AI safety agents without buying new NVR hardware.
- **95%+ Cloud Cost Reduction:** In-browser ONNX Web filtering eliminates 24/7 cloud video streaming bills for small traders.

2. Faster Emergency & Health Triage
- **Sub-5-Second Incident Action:** Replaces next-day DVR review with instant VAPI Hindi phone calls and WhatsApp photo evidence.
- **Hospital & Factory Lives Saved:** Catches OPD patient collapses and unguarded factory machine hazards before injuries escalate.

3. Vernacular Digital Inclusion
- **Zero-Literacy Voice Programming:** Empowers Hindi, Hinglish, and regional speakers to automate complex workflows using natural speech.
- **Active On-Site Deterrence:** Plays immediate regional voice warnings over the local speaker to stop intruders or cattle on the spot.

RESEARCH & REFERENCE

TECHNICAL REFERENCES & REPOSITORY | Grounded in In-Browser ONNX Edge Vision, Multimodal VLM & Voice AI

PRIMARY TECHNICAL & ARCHITECTURE REFERENCES

- **1. In-Browser Edge Vision (ONNX Runtime Web & Ultralytics YOLOv8):**

WebGPU and WebAssembly client-side execution of YOLOv8 object/pose models inside standard browsers with zero server GPU lag (onnxruntime.ai/docs/tutorials/web).

- **2. Multimodal Video & Workflow Reasoning (Gemini 2.5, Gemini Video & Groq):**

Spoken Hindi/Hinglish intent-to-JSON node compilation via Gemini 2.5 and temporal keyframe context verification with Groq LPU inference (ai.google.dev | groq.com).

- **3. Real-World Voice & No-Code Camera Agents (VAPI & Cameron Inspiration):**

Sub-1000ms programmable outbound voice agents in Hindi via VAPI (docs.vapi.ai/providers/model/groq), inspired by no-code browser camera automation tailored for Bharat.

Team Dynamics

- **Lead GitHub Profile:** github.com/farhanmallik05 (CI/CD, Agentic AI & Full-Stack Workspace) for the Submissions
- **In Finale Live Demo:** Speak a Hindi rule on stage → trigger live webcam → Gemini Video verifies → live VAPI Hindi phone call.

Team Empty (5/5):

Farhan Mallik -> Team Lead and presenter
Kartikey Bisht -> ML Specialist
Ankit Maurya -> Backend Developer
Piyush Kumar Gautam -> Frontend Developer
Kaushendra Singh -> AI Developer and tester